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Empathetic Dialogue Generation via Knowledge Enhancing and Emotion Dependency Modeling

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Abstract

Enabling the machines with empathetic abilities to provide context-consistent responses is crucial on both semantic and emotional levels. The task of empathetic dialogue generation is proposed to address this problem. However, two challenges still exist in this task: perceiving nuanced emotions implied in the dialogue context and modelling emotional dependencies. Lacking useful external knowledge makes it challenging to perceive implicit fine-grained emotions. Missing the emotional interactions among interlocutors also restricts the performance of empathetic dialogue generation. To address above challenges, we propose a knowledge-enhanced framework, named Know-EDG. We first enrich dialogue context by bunches of emotion-related concepts and construct a knowledge-enhanced context graph. Then we introduce a graph-aware Transformer encoder to learn graph's semantic and emotional representations, which are the prerequisites of the emotion identifier to predicate the target emotion signal. Finally, we propose an emotion-focused attention mechanism to exploit the emotional dependencies between dialogue context and target empathetic response. Conducted on a benchmark dataset, extensive experimental results show that our proposed framework outperforms state-of-the-art baselines in terms of automatic metrics and human evaluations.

1 Introduction

Studies on social psychology suggest that *empathy* is a crucial factor towards a more humanized dialogue system (Zech and Rimé, 2005). Humans can easily understand the implicit emotions of the conversation partners and display caring attitudes in responses (Rashkin et al., 2019). Although some researchers have attempted to control the emotional content of response either through an explicitly assigned emotional label (Zhou and Wang, 2018; Zhou et al., 2018a; Wang and Wan, 2018; Song et al., 2019; Shen and Feng, 2020) or through a general term to encourage higher levels of affect (Asghar et al., 2018), it is still challenging for chatbots to conduct empathetic dialogue without the explicit emotion labels (Zhou et al., 2018a).

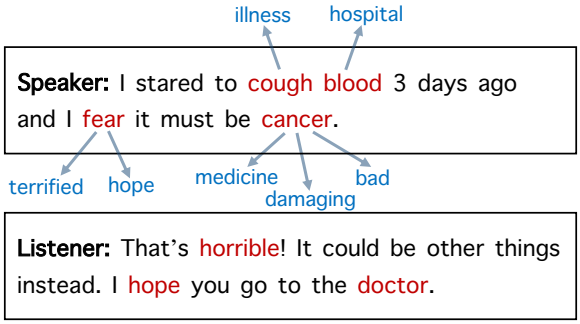

Figure 1: An empathetic dialogue example from dataset EMPATHETICDIALOGUES. The emotional-related words in the dialogue are highlighted in red. The blue words directly connected with red ones are emotion-related concepts.

Several recent works have been proposed to address the empathetic dialogue problem based on multi-task learning (Rashkin et al., 2018, 2019; Wei et al., 2019; Lin et al., 2020) or mixture of experts (Lin et al., 2019). However, an unheeded deep concern is that humans usually rely on experience and external knowledge to reason and express implicit emotions, which inspired us that external knowledge may play a crucial role in emotion information understating and reasoning. From the real example in Figure 1, if use non-stopwords of speaker's input as queries to retrieve external knowledge, we can obtain some emotion-related concepts, "terrified", "hospital", "bad", etc, which could easily help listener to respond in an empathetic way. To illustrate this phenomenon con-

*This work was done when Qintong Li was an intern at Tencent AI Lab.

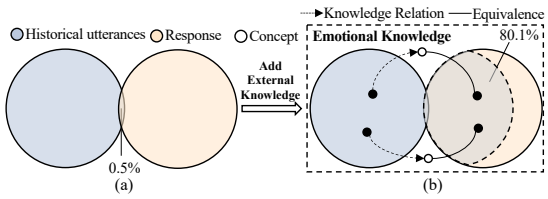

Figure 2: The relationship among historical utterances, responses, situation, and external knowledge.

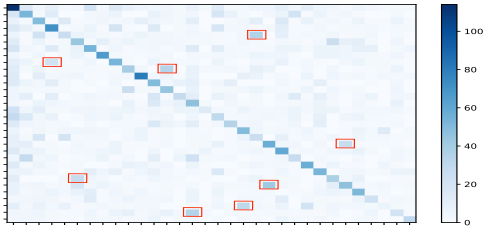

Figure 3: Emotion transition patterns. y-axis indicates the speaker's emotion label. x-axis indicates the listener's emotion label predicted by the classifier.

cretely, we statistically investigate the functions of external knowledge in emotion information understanding on the dataset EMPATHETICDIALOGUES and the conclusions are illustrated in Figure 2.

Figure 2(a) depicts that the response has almost NO non-stopword overlapping (0.5% of dialogue samples) with historical utterances. This phenomenon demonstrates that humans need to reason more information to conduct empathetic dialogues. However, for a chatbot, reasoning emotional knowledge purely based on the provided utterances is tremendously difficult. Then what will happen if we incorporate some emotional knowledge into the system, say retrieve some information (words) from a knowledge base? The answer is depicted in Figure 2(b). We can observe that, in most dialogue samples (80.1%), the chatbot could directly obtain hints from the knowledge paths started by the non-stop tokens in the historical utterances. Therefore, *external knowledge plays an important role in enhancing the performance of empathetic dialogue generation*.

Moreover, during the investigations, we observe an interesting phenomenon that emotional dependency and emotional inertia commonly appear in many real-world conversations. We visualize the emotion transitions from speakers to the listeners in Figure 3. Since the emotion labels for the listener's response are unknown, we only train the CNN-based emotion classifier (Kim, 2014) based on the speaker's utterances and corresponding emotion labels. In Figure 3, the darker diagonal grids show that listeners tend to mirror the emotion of their interlocutors to build rapport (Navarretta, 2016). Moreover, there are also some complex emotional transition patterns besides the diagonal direction (in red frame). Therefore, intuitively, modelling emotional dependencies between interlocutors is crucial and helpful to improve the empathetic dialogue performance.

To handle the pre-mentioned issues, we propose a Knowledge-enhanced framework, **Know-EDG**, for empathetic dialogue generation. Know-EDG consists of three components: (1) *Knowledge-enhanced context encoder* for conducting external-knowledge acquirement; (2) *emotion identifier* for jointly modeling the dynamic emotion transition mechanism; (3) *emotion-focused response generator* for generating the empathetic response by jointly considering the emotional information in dialogue context and the predicted ones. Conducted on the benchmark dataset EMPATHETIC-DIALOGUES (Rashkin et al., 2019), extensive experimental results demonstrate the effectiveness of Know-EDG in terms of both automatic metrics and human evaluation.

In summary, our contributions are as follows: (1) We are the early attempt that propose a knowledge-enhanced framework (Know-EDG) to accurately perceive the implicit emotions in empathetic dialogue generation. (2) The obtained external knowledge as well as the utterances are constructed into a graph to conduct the semantic and emotional feature representation learning. (3) An emotion identifier and emotion-focused attention mechanism are proposed to capture the dynamic emotion transition information and predicate the emotion signal for the target response. (4) Experimental results show the proposed approach outperforms competitive baselines in terms of both automatic evaluation metrics and human evaluations.

2 Related Work

Emotional Dialogue Generation: With the rise of data-driven learning approaches (Sutskever et al., 2014; Vaswani et al., 2017), open-domain dialogue generation models have seen growing interests in recent years (Vinyals and Le, 2015; Shang et al., 2015; Serban et al., 2016; Li et al., 2016; Zhou et al., 2018b; Dinan et al., 2019). To control the emotional content of the target output, recent approaches generate emotional responses condition-

ing on a manually specified label (Zhou et al., 2018a; Li and Sun, 2018; Zhou and Wang, 2018; Huang et al., 2018; Colombo et al., 2019; Shen and Feng, 2020). However, existing emotional dialogue models purely focus on whether the generated response matches a predetermined emotion, whereas in real-world scenarios the listener is capable to infer the emotion of the speaker (Rashkin et al., 2019).

Empathetic Dialogue Generation: Unlike the task of emotional dialogue generation, the task of empathetic dialogue generation avoids an additional step of determining which emotion type to respond explicitly (Skowron et al., 2013). Several works (Lubis et al., 2018; Rashkin et al., 2018; Zhong et al., 2019a; Wei et al., 2019; Shin et al., 2019; Chatterjee et al., 2019; Rashkin et al., 2019; Santhanam and Shaikh, 2019; Lin et al., 2019, 2020; Zhong et al., 2020) have attempted to make dialogue models more empathetic. Rashkin et al. (2019) combined existing models in different ways to produce empathetic responses. Lin et al. (2019) softly combined the possible emotional responses from several separate experts to generate the final empathetic response. Besides the advancements in empathetic dialogue models, the emergence of new emotion-labelled dialogue corpora have also contributed to this research field. Li et al. (2017) introduced DAILYDIALOG dataset, with manually emotion labelling to each utterance in the multi-turn dialogue setting. Hsu et al. (2018) collected a multi-party dataset EMOTIONLINES where each utterance is labelled with one of seven emotions. However, only 5% of the utterances in DAILYDIALOG and 16.68% of the utterances in EMOTIONLINES have diverse emotional labels and others are either none or happy labels. Because of the extremely imbalanced data distribution, they are not suitable to be engaged as the benchmarks of empathetic dialogue generation. Rashkin et al. (2019) considered a richer and evenly distributed set of emotions and release a dataset EMPATHETICDIALOGUES, where a listener responds to a speaker who is under an emotional situation in an empathetic way. In out work, we focus on the task of empathetic dialogue generation on the dataset of EMPATHETICDIALOGUES.

3 Preliminaries

In this section, we introduce the external knowledge used in Know-EDG: the commonsense knowledge in ConceptNet (Speer et al., 2017) and the emotional knowledge NRC_VAD (Mohammad, 2018).

ConceptNet is a large-scale knowledge graph that describes general human knowledge in natural language. It comprises 5.9M tuples, 3.1M concepts, and 38 relations. We denote each tuple (head concept, relation, tail concept, confidence score) as $\tau = (h, r, t, s)$ where the confidence score $s \in [1, 10]$. We use *min-max* normalization to scale s between 0 and 1:

$$\textit{min-max}(s) = \frac{s - \min_s}{\max_s - \min_s}, \quad (1)$$

where $\min_s$ and $\max_s$ is 1 and 10, respectively. For example, the confidence score s in tuple $\langle \text{birthday}, \text{RelatedTo}, \text{happy}, s \rangle$ is normalized from 2.69 to 0.19.

NRC_VAD is a list of VAD (Valence-Arousal-Dominance) vectors with dimensions (V_a, A_r, D_o) for 20k English words, which are culture-independent and widely adopted in Psychology (Mehrabian, 1996). The interpretations of VAD vectors are presented in Table 1. For example, the VAD vector of word "nice" is: $[0.93, 0.442, 0.65]$.

Dimensions	Values	Interpretations
Valence	$[0, 1]$	Negative - Positive
Arousal	$[0, 1]$	Calm - Excited
Dominance	$[0, 1]$	Submissive - Dominant

Table 1: Interpretations of NRC_VAD vectors.

Inspired by Zhong et al. (2019b), we adopt NRC_VAD to compute the emotion intensity for words in dialogue context and concepts from ConceptNet.

$$\eta_i = \textit{min-max}(\left\| V_a(x_i) - \frac{1}{2}, \frac{A_r(x_i)}{2} \right\|_2), \quad (2)$$

where *min-max* is the min-max normalization; $\|.\|_k$ denotes L_k norm; $V_a(x_i)$ and $A_r(x_i)$ denote valence and arousal dimension values in VAD vector of word x_i, respectively. If x_i not in NRC_VAD, $V_a(x_i)$ and $A_r(x_i)$ will be set to 0.5.

4 Model

We propose to use commonsense- and emotion-knowledge sources for an open-ended empathetic conversational system to ground in.

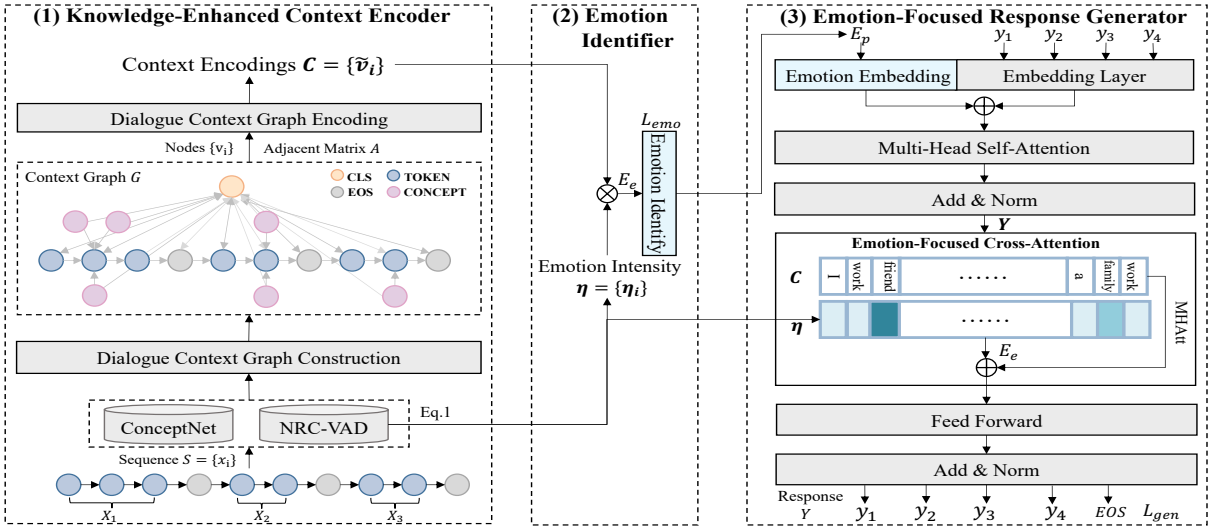

Figure 4: An overall architecture of our Know-EDG model.

4.1 Overview

Formally, we are given a dialogue history of turns $[X_1, \ldots, X_M]$ where X_i is the i-th utterance, the commonsense knowledge graph ConceptNet, and the list of VAD vectors VADs. The system needs to generate an empathetic response $Y = \{y_1, \ldots, y_n\}$ that is both emotionally appropriate and meaningful with the content.

The proposed framework Know-EDG is shown in Figure 4. Know-EDG consists of three components. (1) **Knowledge-Enhanced Context Encoder**: We construct a knowledge-enriched context graph $G = (V, A)$ as the dialogue context, where V is a set of nodes (a root node CLS, utterance token nodes, and knowledge concept nodes) and A denotes an adjacency matrix describing the directed edges (See §4.2.1). The dialogue context is then encoded into vector representations by a graph-aware Transformer encoder (Koncel-Kedziorski et al., 2019) (See §4.2.2). (2) **Emotion Identifier**: We distill the emotion information from the dialogue context and predicate the emotion signal for the response generation (See §4.3). (3) **Emotion-Focused Response Generator**: We design an emotion-focused attention mechanism to learn the emotional dependencies from the dialogue context and generate the final response (See §4.4).

4.2 Knowledge-Enhanced Context Encoder

We first describe the construction procedure of the knowledge-enriched context graph. Then we introduce the encoding details of the dialogue context.

4.2.1 Dialogue Context Graph Construction

To enrich the dialogue context with the emotion-related knowledge, we construct a context graph in the preprocessing stage.

Following Lin et al. (2019), we flat M utterances into a long token sequence S. For each non-stopword token x_i in S, we retrieve a set of candidate tuples $\{\tau_k^i = (x_i, r_k^i, c_k^i, s_k^i)\}_{k=1,\ldots,K}$. To refine the emotion-related knowledge from the commonsense knowledge, we filter the retrieved tuples in two steps:

- **Relation Filtering**: Since r_k^i in ConceptNet are diversified and some are irrelevant, *e.g.*, "NotHasProperty", we create an excluded relation list L_{ex} to filter the set of retrieved tuples and reserve the tuples whose $r_k^i \notin L_{ex}$ and $s_k^i > \alpha$. α is a pre-defined threshold the same as the one in Zhong et al. (2019b).

- **Concept Ranking**: We first derive a matching score for each tuple τ_k^i according to three aspects (emotion, semantic, and confidence):

$$\text{Score}\left(\tau_k^i\right) = f_e(c_k^i) + \cos(x_i, c_k^i) + s_k^i,$$
$$f_e(c_k^i) = \textit{min-max}(\|V_a(c_k^i) - \frac{1}{2}, \frac{A_r(c_k^i)}{2}\|_2), \quad (3)$$

where $f_e(c_k^i)$ is the emotion intensity of tail concept c_k^i, *min-max* is the min-max normalization function, $\cos()$ is the cosine similarity function between tokens. Then we select top K' tuples $\{\tau_k^i = (x^i, r_k^i, c_k^i, s_k^i)\}_{k=1,\ldots,K'}$ as the emotional knowledge for each token x_i.

For the context graph $G = (V, A)$, both the tokens x_i ($i = 1, \ldots, |S|$) in dialogue history and the tail concepts c_k^i ($i = 1, \ldots, |S|$; $k = 1, \ldots, K'$) in emotional knowledge form $V = \{v_i\}_{i=1,\ldots,m}$, where m is the number of nodes. G contains 3 directed relation types: (a) *sequence*: forward relation between two successive tokens, i.e., $x_i \rightarrow x_{i+1}$. (b) *emotion*: relation between the token x_i and its concepts c_k^i, i.e., $c_k^i \rightarrow x_i$. (c) *globality*: relation between CLS node and other nodes, i.e. $x_i \rightarrow$ CLS, $c_k^i \rightarrow$ CLS, and CLS $\rightarrow x_i$. All the three directed relations among nodes are set to 1 in the adjacency matrix A.

4.2.2 Dialogue Context Graph Encoding

We first use a word embedding layer and a positional embedding layer (Vaswani et al., 2017) to convert each node v_i into vectors $\mathbf{E}_w(v_i) \in \mathbb{R}^d$ and $\mathbf{E}_p(v_i) \in \mathbb{R}^d$, where d is the dimensionality of embeddings. In the multi-turn dialogue settings, distinguishing nodes in different utterances is helpful. So we incorporate the dialogue state embedding $\mathbf{E}_d(v_i)$ for node v_i. The vector representation of node v_i is the composition of three types of embeddings:

$$\mathbf{v}_i = \mathbf{E}_w(v_i) + \mathbf{E}_d(v_i) + \mathbf{E}_p(v_i). \tag{4}$$

Then we apply a multi-head graph-attention mechanism to update the node representations with emotional knowledge. Specifically, each node $\mathbf{v}_i$ is contextualized by attending to all its connected neighbour vertices $\{\mathbf{v}_j\}$.

$$\begin{aligned} \hat{\mathbf{v}}_i &= \mathbf{v}_i + \mathop{\Big\|}_{n=1}^{H} \sum_{j \in \mathcal{A}_i} \alpha_{ij}^n \mathbf{W}_v^n \mathbf{v}_j, \\ \alpha_{ij}^n &= a^n(\mathbf{v}_i, \mathbf{v}_j). \end{aligned} \tag{5}$$

Here, $\|$ denotes the concatenation of H attention heads, $\mathcal{A}_i$ denotes the neighborhood of v_i in the adjacency matrix A, function $a^n()$ represent the self-attention mechanism of the n-th head in the following format:

$$a^n(\mathbf{q}_i, \mathbf{k}_j) = \frac{\exp((\mathbf{W}_q^n \mathbf{q}_i)^\top \mathbf{W}_k^n \mathbf{k}_j)}{\sum_{z \in \mathcal{A}_i} \exp((\mathbf{W}_k^n \mathbf{k}_z)^\top \mathbf{W}_q^n \mathbf{q}_i)}, \tag{6}$$

where $\mathbf{W}_q^n$, $\mathbf{W}_k^n$ are the linear transformations of the node $\mathbf{q}$ and node $\mathbf{k}$ respectively.

Note that the previous operations are only conducted to the local context (neighbours), thus we update the node representations with the global contextual information to model the node interactions among different utterances. Precisely, we flat all the dialogue utterance nodes into a long sequence by their temporal order and append the concept nodes to the end in descending order according to the corresponding emotion intensity. Then we use the l-th transformer layer (Vaswani et al., 2017) to encode global information for all nodes $\{\hat{\mathbf{v}}_i\}_{i=1,\ldots,m}$.

$$\mathbf{h}_i^l = \text{LayerNorm}(\hat{\mathbf{v}}_i^{l-1} + \text{MHAtt}(\hat{\mathbf{v}}_i^{l-1})), \tag{7}$$

$$\tilde{\mathbf{v}}_i^l = \text{LayerNorm}(\mathbf{h}_i^l + \text{FFN}(\mathbf{h}_i^l)), \tag{8}$$

where LayerNorm is the Layer Normalization trick proposed in (Ba et al., 2016); MHAtt is the multi-head self-attention sub-layer consiting of H attention heads; FFN is a two-layer feed-forward network with ReLU as hidden activation function. The transformer layers are stacked L times. The obtained final context representations are denoted as $\mathbf{C} = \{\tilde{\mathbf{v}}_i\}_{i=1,\ldots,m}$.

4.3 Emotion Identifier

This component is designed to conduct the emotion information predicting for the target response. We utilize Eq. 2 to calculate the emotion intensity values $\{\eta_i\}$ for all nodes $\{\tilde{\mathbf{v}}_i\}$. By treating $\{\eta_i\}$ as the node emotion information, we derive the emotion-focused context representation $\mathbf{E}_e$ by the weighted summation of all the node representations $\{\tilde{\mathbf{v}}_i\}$:

$$\mathbf{E}_e = \sum_i \frac{\exp(\eta_i)}{\sum_j \exp(\eta_j)} \tilde{\mathbf{v}}_i. \tag{9}$$

Then a linear layer with $\mathrm{softmax}$ operation projects the emotion-enhanced context $\mathbf{E}_e$ into an emotion category distribution $\mathbf{e}_p$ to identify the emotion signal for the target response:

$$\mathbf{E}_p = \mathbf{W}_e \mathbf{E}_e, \tag{10}$$

$$\mathbf{e}_p = \text{softmax}(\mathbf{E}_p), \tag{11}$$

where $\mathbf{W}_e \in \mathbb{R}^{d \times q}$ and q is the number of emotion categories. During training, given the golden truth one-hot emotion label $\mathbf{e} \in \mathbb{R}^q$, we employ cross-entropy as the loss function to conduct the parameter learning:

$$\mathcal{L}_{emo} = -\sum_i \mathbf{e}^i \log(\mathbf{e}_p^i). \tag{12}$$

Moreover, the obtained intermediate emotion-enhanced context representation $\mathbf{E}_p$ will be fed into the decoder as a crucial emotional feature to guild the empathetic response generation.

4.4 Emotion-Focused Response Generator

The emotion signal $\mathbf{E}_p \in \mathbb{R}^{1\times q}$ is firstly be transformed by a linear transformation into $\mathbf{E}'_p \in \mathbb{R}^{1\times d}$. Then we concatenate $\mathbf{E}'_p$ with the embeddings of the decoder input tokens $y_{1:j-1}$ into representations $\mathbf{Y}_{emb} = \{\mathbf{y}_t\}_{t=0,\ldots,j-1}$ where $\mathbf{y}_0 = \mathbf{E}'_p$. We feed $\mathbf{Y}_{emb}$ into the response generator.

The generator is built based on L Transformer layers as well. For each Transformer decoder layer, the decoder inputs $\mathbf{Y}_{emb}$ are converted into new vector representations $\mathbf{Y} = \{\mathbf{y}_t\}_{t=0,\ldots,j-1}$ by a Transformer decoder layer.

To learn the emotional dependencies between the dialogue context $\mathbf{C}$ and the predicted response $\mathbf{Y}$ and generate an empathetic response with appropriate emotions, an **emotion-focused cross attention mechanism** E-MHAtt is designed to replace the cross-attention sub-layer in the original Transformer decoder layer. The intuition is that humans can naturally pay extra attention to the emotional salient information during the conversation. With the emotion dependency vectors produced by E-MHAtt, the representations of the predicted response are enhanced as follows:

$$\mathbf{D} = \mathbf{Y} + \mathbf{W}_m \text{E-MHAtt}(\mathbf{Y}, \mathbf{C}), \tag{13}$$

where E-MHAtt is the concatenation of the context vector from multi-head attention MHAtt (Vaswani et al., 2017) and emotion vector $\mathbf{E}_e$:

$$\text{E-MHAtt}(\mathbf{Y}, \mathbf{C}) = [\text{MHAtt}(\mathbf{Y}, \mathbf{C}) \| \mathbf{E}_e]. \tag{14}$$

Similar with the Transformer decoder, the remaining operations are as following:

$$\hat{\mathbf{D}} = \text{LayerNorm}(\mathbf{D}), \tag{15}$$
$$\hat{\mathbf{Y}} = \text{LayerNorm}(\hat{\mathbf{D}} + \text{FFN}(\hat{\mathbf{D}})), \tag{16}$$

where $\hat{\mathbf{Y}} = [\hat{\mathbf{y}}_1, \ldots, \hat{\mathbf{y}}_j]$. Then the response generator yeilds the distribution over the vocabulary for the next j-th token:

$$\alpha^g(y_j|\mathbf{C}, y_{0:j-1}) = \text{softmax}(\mathbf{W}_o \hat{\mathbf{y}}_\mathbf{j}). \tag{17}$$

Since external concepts can introduce the emotional knowledge for empathetic responding, we compute a probability p_g of copying from nodes $\{v_i\}_{i=1,\ldots,m}$ in the context graph in a manner similar to See et al. (2017) and derive the final next-token probability distribution $p(y_j)$:

$$p_g = \sigma(\mathbf{W}_c \hat{\mathbf{y}}_j + b_c), \tag{18}$$
$$p(y_j) = (1 - p_g) * \alpha^c + p_g * \alpha^g, \tag{19}$$

where the copy probability α^c over the knowledge-enriched dialogue context is obtained from the concatenation of attention weights $\{\alpha^c_i = \alpha(\hat{\mathbf{y}}_j, \tilde{\mathbf{v}}_i)$, for $\tilde{\mathbf{v}}_i \in \mathbf{C}\}$. The computation formation of function α is same in (See et al., 2017).

As most dialogue generation tasks, we use standard maximum likelihood estimator (MLE) as the optimization objective:

$$\mathcal{L}_{gen} = -\log p(y_j|y_{<j}, \mathbf{C}). \tag{20}$$

Finally, considering all the aforementioned components, we define a joint loss function as:

$$\mathcal{L}_{gen} = \gamma_1 \mathcal{L}_{emo} + \gamma_2 \mathcal{L}_{gen}, \tag{21}$$

where γ_1, γ_2 are hyper-parameters that control the weights of the two losses. We set $\gamma_1 = \gamma_2 = 1$. All the parameters are jointly trained in an end-to-end paradigm.

5 Experimental Settings

5.1 Data Preparation

We conduct our experiments on the EMPATHETIC-DIALOGUES dataset (Rashkin et al., 2019). EMPATHETICDIALOGUES is a large-scale multi-turn empathetic dialogue dataset collected on the Amazon Mechanical Turk, containing about 25k one-to-one open-domain conversation. Specifically, Rashkin et al. (2019) pair two crowd-workers: a speaker and a listener. The speaker is asked to talk about the personal emotional feelings. The listener infers the underlying emotion through what the speaker says and responds empathetically. The dataset provides 32 evenly distributed emotion labels. At training time, the emotional label of the historical utterances (*i.e.*, the speaker) acts as a supervised signal, while we hide the label in test time to evaluate the empathetic ability of all the models. We treat the historical utterances as the system input and the listener's response as the target output. Then we obtain 17,802 dialogues in the training set, 2,628 in the validation set, and 2,494 in the testing set. The average historical utterances and response lengths are 2.1 utterances and 13.5 tokens respectively.

5.2 Baselines

We compare with the state-of-the-art baselines as follows:

- **Transformer** (Vaswani et al., 2017): A Transformer-based sequence to sequence model that is trained based on MLE loss.

Models	Accuracy	Perplexity	Distinct-1	Distinct-2	Empathy	Relevance	Fluency
Transformer	-	37.73	0.47*	2.04*	3.11	3.47	3.66
EmoPrepend-1	0.3452	38.30	0.46*	2.08*	3.23	3.51	**3.74**
MoEL	0.3524	38.04	0.44*	2.10*	3.37	3.78	3.64
Know-EDG	**0.3931**	**34.85**	**1.48**	**4.90**	**3.49**	**3.91**	3.65

Table 2: Evaluation results of baselines and our models. **Bold face** indicates leading results in terms of the corresponding metric. t-test is conducted between Know-EDG and baselines. (underline: *p-value* < 0.05, *: p-value < 0.01)

- **EmoPrepend-1** (Rashkin et al., 2019): An extension of the Transformer model which incorporates an additional supervised emotion classifier. The whole model is jointly trained by optimizing both the classification and generation loss.

- **MoEL** (Lin et al., 2019): Another extension of Transformer model which softly combines the response representations from different transformer decoders. Each decoder is optimized to focus on one type of emotion accordingly.

Additionally, we also conduct ablation studies to better analyze the influence of different components in our model:

- **w/o CE**: The Know-EDG model without the Knowledge-Enhanced Context Encoder.

- **w/o EFAM**: The Know-EDG model without the Emotion-Focused Attention Mechanism.

5.3 Implementation Details

We lowercase the characters, tokenize the sequences and retain a vocabulary with 24,646 tokens. We use pre-trained Glove vectors (Pennington et al., 2014) to initialize the word embedding. All common hyperparameters are the same as the work in (Lin et al., 2019). The maximum introducing numbers of external concepts per dialogue and per token are set as 10 and 5, respectively. The threshold α used in dialogue context graph construction is 0.1. We implemented all models in PyTorch (Paszke et al., 2017) with a single Tesla V100 GPU, and train models using Adam optimization (Kingma and Ba, 2014) with a mini-batch size of 16. The parameters of the Transformer, Emoprepend_1, MoEL, and our model Know_EDG are 16M, 16M, 21M, and 31M. We varied the learning rate during training following Vaswani et al. (2017). Early stopping is applied when training. The training time of model Know_EDG is 3 hours for around 21000 iterations. When inference, we set the maximum decoding step as 30.

5.4 Evaluation Metrics

Automatic Evaluations To evaluate the model at the emotional level, we adopt *Emotion Accuracy* as the agreement between the ground truth emotion labels and the predicted emotion labels by the emotion identifier. Liu et al. (2016) have verified BLEU is not suitable for measuring dialogue generation quality due to its low correlation with human judgment; METEOR (Banerjee and Lavie, 2005) and ROUGE (Lin, 2004) have the same problem. Therefore, following previous emotion-related studies (Zhou et al., 2018a; Rashkin et al., 2019; Song et al., 2019; Wei et al., 2019), we adopt Perplexity (Serban et al., 2015), Distinct-1, and Distinct-2 (Li et al., 2014) to evaluate comparisons in our experiments: Perplexity measures the high-level general quality of the generation model. Distinct-1 / Distinct-2 is the proportion of the distinct unigrams / bigrams in all the generated results to indicate the diversity.

Human Evaluations. We randomly sample 100 dialogues and their corresponding generations from our model as well as the baselines. We recruit three professional annotators from a third-party company to evaluate the responses generated by different models. All models are evaluated in terms of following 3 metrics: Empathy, Relevance and Fluency (Rashkin et al., 2019; Lin et al., 2019). Empathy measures whether the listener's responses show the understanding of the speaker's feelings; Relevance evaluates whether the generated responses are on-topic with the historical utterances; Fluency measures the grammatical correctness and readability of the generated responses. Each metric is rated on five-scale, where 1, 3, and 5 indicate unacceptable, moderate, and excellent performance, respectively.

Models	Accuracy	Perplexity	Distinct-1	Distinct-2
Know-EDG	**0.3931**	**34.85**	1.48	4.90
w/o CE	0.3637	35.52	**1.55**	**5.61**
w/o EFAM	0.3802	35.48	0.95	2.69

Table 3: Ablation study.

Models	Win	Loss	Tie
Know-EDG vs Transformer	43.8%	17.5%	38.7%
Know-EDG vs EmoP	38.3%	18.0%	43.7%
Know-EDG vs MoEL	36.6%	20.6%	42.8%

Table 4: Result of human A/B test. Tests are conducted pairwise between Know-EDG and baseline models.

6 Results and Analysis

We show both automatic and human evaluation results in Table 2.

Automatic Evaluation Results We observe that our model Know-EDG outperforms state-of-the-art baseline MoEL by a large margin in terms of all automatic metrics. The noticeable improvement indicates the effectiveness of our knowledge-enhanced framework in empathetic expression and response diversity. EmpPrepend-1 and MoEL have similar performance, as both of them only use the historical utterances to infer emotional states and generate responses. Without emotion modelling, Transformer only generates fluent responses based on semantic mapping, but fail to express diverse responses.

We also perform an ablation study for better understanding the contributions of the main parts of our model. As shown in Table 3, after we remove the Knowledge-Enhanced Context Encoder (w/o CE model), both the emotion accuracy and perplexity performance become obviously worse, indicating that injecting external knowledge is consistently critical for emotion understanding and model generation quality. We also investigate the effect of removing Emotion-Focused Attention Mechanism (i.e., w/o EFAM model). We notice that the scores decrease dramatically on all metrics, which demonstrates the effectiveness of the Emotion-Focused Attention Mechanism in modelling emotional dependencies.

Human Evaluation Results Table 2 illustrates that Know-EDG obtains the best performance on both Empathy and Relevance scores. This suggests that the emotion-focused attention mechanism on knowledge-enriched dialogue context helps to capture implicit emotions, improve the topic consistency, and elicits a more appropriate response. We see there is no obvious difference among models in terms of Fluency. We deduce it's because the generated responses by Transformer are already fluent and grammatical. Additionally, we carried out pairwise response comparison to directly compare the dialogue quality gains in Table 4. The results confirm that the responses from Know-EDG are more preferred by human judges.

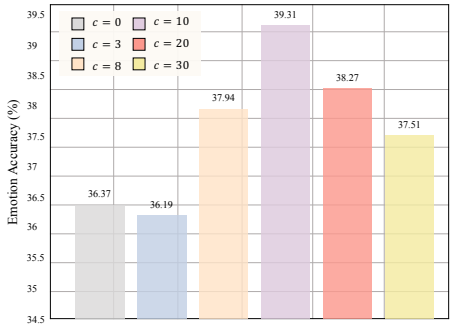

Figure 5: Emotion accuracy with respect to the number of external concepts. $c = 8$ indicates at most we introduce 8 external concepts for each dialogue.

Utterance	It inspires me to try and do something to keep healthy every day .
MoEL	I am sure they will be able to have a good time.
Utterance	It inspires me to try and do something to keep healthy every day .
Knowledge	effort , fight , good , life , raise , grow , protect , health
Know-EDG	I can not wait to try to get a little makes me feel better.

Table 5: The visualization of the cross-attention weights in MoEL and Know-EDG.

6.1 External Knowledge Analysis

To further investigate the impact of external knowledge, we train our model with different number of concepts and evaluate by Accuracy as shown in Figure 5. With increasing the number of concepts, the performance is rising. However, if we introduce too many concepts, the accuracy no longer increase or even decrease. Therefore, external knowledge is more suitable to be the auxiliary information to understand the emotional states in the historical utterances.

Emo. Cont.	**Terrified** X_1: Do you know how crazy it is to skydive? X_2: I have a fear of falling from high places. X_3: It gave me the biggest rush that is for sure.
Gold	**I think I would pass out from fear lol.**
Transfmr	I am sure it was.
EmoP	I am sure it was a lot of fun.
MoEL	I am sure you will do it!
Know.	danger, pain, travel, scared, excited, furious
EDG	I would have been so scared.
Emo. Cont.	**Surprised** X_1: I realized yesterday I was pregnant, I was in shock!
Gold	**Woah, that is huge news. How are you handling the news.**
Transfmr	Oh no! I am so sorry to hear that.
EmoP	Oh no! Did you get a job?
MoEL	That is so awesome! I am glad to hear that.
Know.	experience, surprise, mother, pain, feeling
EDG	Oh wow ! that is a great feeling.

Table 6: Generated responses from Transformer (short in Transfmr), EmoPrepend-1 (short in EmoP), MoEL, and Know-EDG (short in EDG) in two different speaker's emotion states. Tokens in underline represent knowledge-related words.

6.2 Emotion-Focused Attention Analysis

Table 5 shows an example illustrating the cross-attention weights of the dialogue context. Baseline MoEL put the major attention on the general words, which leads to a context-inconsistent and emotion-inappropriate response. In comparison, the Know-EDG model puts the highest attention probability on the words containing informative meaning, e.g., "fight" and "grow" in external knowledge and "keep" and "healthy" in historical utterances. We can conclude that the proposed emotion-focused attention mechanism can teach the model to generate responses from meaningful and emotional words.

6.3 Case Study

Some cases generated by Know-EDG and the baseline models are listed in Table 6. In the first case, Know-EDG generates both coherent and informative responses with a proper negative emotion by replying with "scared". But all baselines fail to understand that the emotion of the utterances is negative, i.e., terrified, the baselines tend to express positive emotions without emotional knowledge. In the second case, Know-EDG model generates the most context-consisten response, which contains context-related word ("feeling") and emotion-rated word ("Oh wow"). Both the two cases show that the Know-EDG model can balance the performances between content and emotion.

7 Conclusion

We propose a knowledge-enhanced framework, Know-EDG, to enhance the performance of empathetic dialogue generation. We enrich the dialogue utterances by introducing some external concepts and construct a dialogue context graph. Then we employ a graph-aware Transformer encoder to learn graph's semantic and emotional representations. Finally, we design an emotion-focused attention mechanism to exploit the emotional dependencies between the dialogue context and the target empathetic response. Experimental results show that our framework outperforms all state-of-the-art baselines in terms of automatic metrics and human evaluations.

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